

AI VIET NAM

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SQL in Data Analysis

(Introduction to Database)

Vinh Dinh Nguyen
PhD in Computer Science

Outline



- **Introduction to Database**
- **How to install MySQL**
- **How to create Database**
- **How to use SQL for data analysis**
- **Kahoot Quiz**
- **Summary**



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Introduction to Database

Database Motivation

Data for Online Shopping System

Customer ID	Name	Address	Product Name	Quantity	Date
1	Nguyen Van A	Tay Ninh	Monitor	5	01-01-2024
1	Nguyen Van A	Tay Ninh	Mouse	6	01-01-2024
2	Nguyen Van B	Can Tho	Monitor	4	01-01-2024
2	Nguyen Van B	Can Tho	Keyboard	5	02-01-2024
3	Nguyen Thi C	TP. HCM	Keyboard	5	02-01-2024
3	Nguyen Thi C	TP. HCM	Mouse	4	03-01-2024



What's a problem for a hundred thousand customers?



Database Motivation

Customer

C_ID	Name	Address
1	Nguyen Van A	Tay Ninh
2	Nguyen Van B	Can Tho
3	Nguyen Thi C	TP. HCM

Product

P_ID	Name	Quantity
1	Monitor	100
2	Mouse	100
3	Keyboard	100

↓
Index
Primary Key

Values

Solution

Order

C_ID	P_ID	Quantity	Date
1	1	5	01-01-2024
1	2	6	01-01-2024
2	1	4	01-01-2024

↓
Index/
Foreign Key

Values

Database Motivation

Customer

C_ID	Name	Address
1	Nguyen Van A	Tay Ninh
2	Nguyen Van B	Can Tho
3	Nguyen Thi C	TP. HCM

Product

P_ID	Name	Quantity
1	Monitor	100
2	Mouse	100
3	Keyboard	100

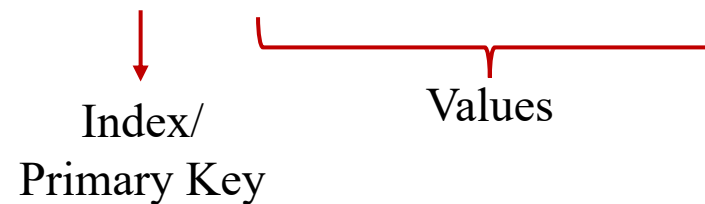


 Index
Primary Key

Order

Another Solution

O_ID	C_ID	Date
1	1	01-01-2024
2	2	01-01-2024



 Index/
 Primary Key

Order Detail

OD_ID	O_ID	P_ID	Quantity
1	1	1	5
2	1	2	6
3		1	4



 Index/
 Primary Key

Definition

A collection of information/data.

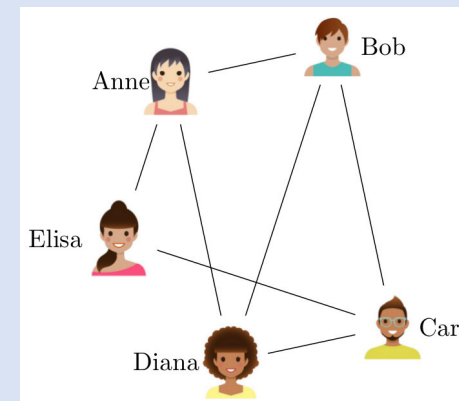
Stored and organized in some format.

Easily accessed, managed and retrieved

Databases



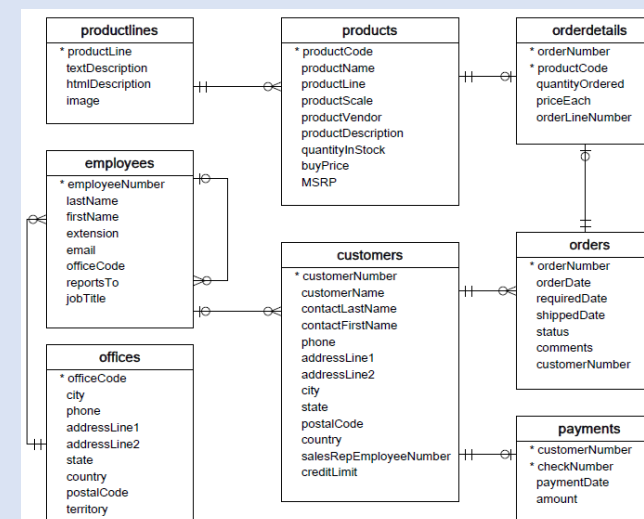
Phone book



Friend Graph

1. DAIRY Milk Eggs Butter Cheese Yogurt	6. CONDIMENTS/PRESERVES Salt Pepper Cooking oil Vinegar Honey Stock cubes
2. MEAT/PROTEIN Chicken/poultry Red meat White fish Oily fish Charcuterie Tofu	Herbs: Mixed herbs Basil Oregano Coriander
3. PRODUCE Potatoes Carrots Onions Garlic	Spices: Chilli powder Paprika Cumin Cinnamon Ginger Nutmeg
4. CANNED & DRIED PRODUCE Chopped tomatoes Soup Pulses Fruit, nuts, and seeds	7. FREEZER Peas Mixed vegetables Fish fingers
5. BAKING Sugar Flour: self-raising & plain Baking powder Vanilla extract	8. BREAD/PASTA/GRAINS Pasta Rice Bread Breakfast cereal

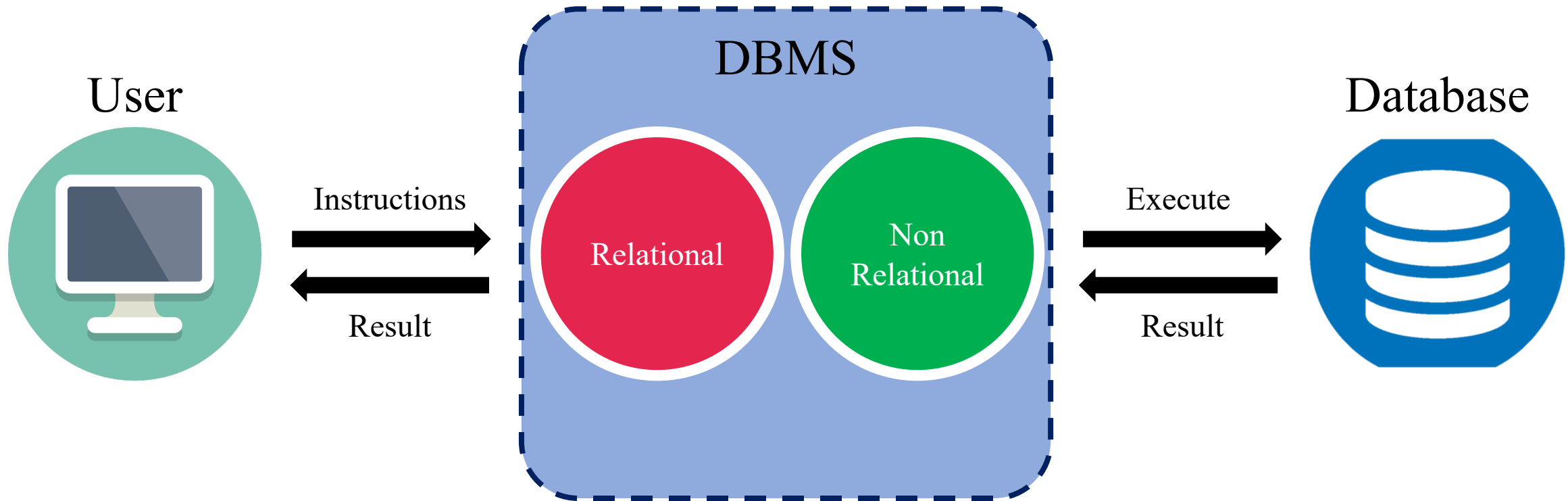
Shopping list



Relation database

Database Management System

A software or application that allow user to interact with the database.



Relational Databases

RELATIONAL DATABASES

Customers

Products

Orders

STRUCTURED QUERY LANGUAGE (SQL)

```
SELECT *  
FROM products  
WHERE category = 'food'  
ORDER BY price
```

C.R.U.D

Create Read Update Delete

RELATIONAL DBMS



ORACLE®
DATABASE

Outline



➤ **Introduction to Database**

➤ **How to install MySQL**

➤ **How to create Database**

➤ **How to use SQL for data analysis**

➤ **Kahoot Quiz**

➤ **Summary**



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Installing MySQL



Installing MySQL on MAC



Step 1: Go to mysql.com.

Step 2: Click on the **DOWNLOADS** tab

Step 3: Scroll to the bottom of the pages and click [MySQL Community \(GPL\) Downloads »](#)

Step 4: Click on **MySQL Community Server** and Download the first link

MySQL Community Downloads

- MySQL Yum Repository
- MySQL APT Repository
- MySQL SUSE Repository
- MySQL Community Server**
- MySQL Cluster
- MySQL Router
- MySQL Shell
- MySQL Operator
- MySQL NDB Operator
- MySQL Workbench
- MySQL Installer for Windows
- C API (libmysqlclient)
- Connector/C++
- Connector/J
- Connector/NET
- Connector/Node.js
- Connector/ODBC
- Connector/Python
- MySQL Native Driver for PHP
- MySQL Benchmark Tool
- Time zone description tables
- Download Archives

General Availability (GA) Releases Archives ⓘ

MySQL Community Server 8.0.33

Select Operating System:
macOS ⓘ

Select OS Version:
All ⓘ

Looking for previous GA versions?

ⓘ Packages for Ventura (13) are compatible with Monterey (12)

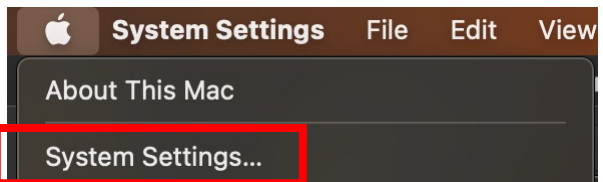
macOS 13 (ARM, 64-bit), DMG Archive (mysql-8.0.33-macos13-arm64.dmg)	8.0.33	559.1M	Download
MD5: 065662283bf45231ea0b47e4ab5f5eb37 Signature			
macOS 13 (x86, 64-bit), DMG Archive (mysql-8.0.33-macos13-x86_64.dmg)	8.0.33	564.5M	Download
MD5: ddf70a939f0b64d585393e3b3fd4fe36c Signature			

Installing MySQL on MAC

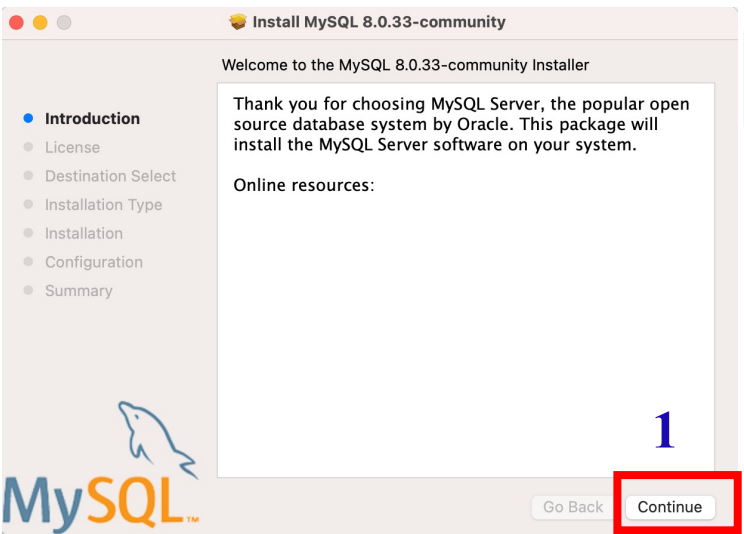
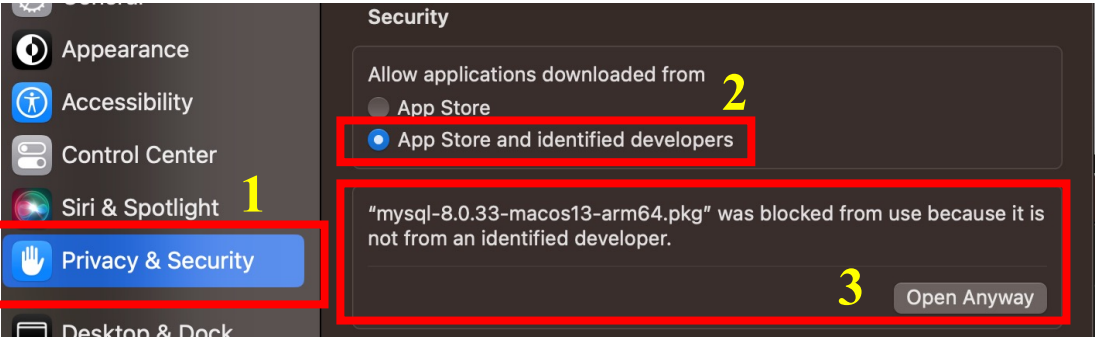
Step 5: Open the download file.



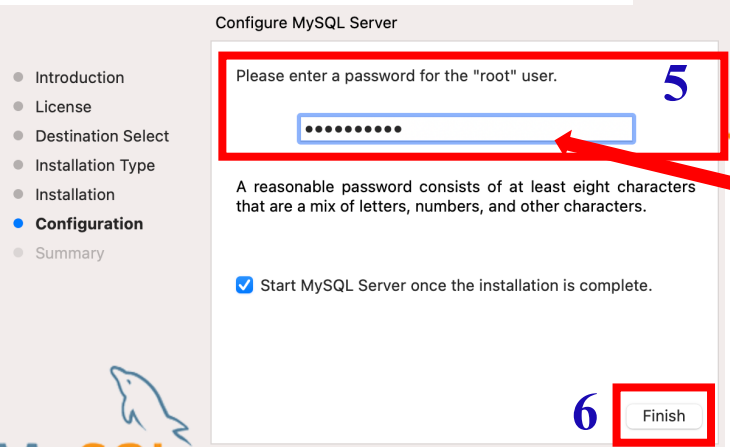
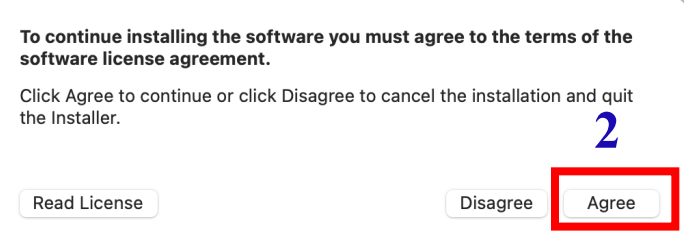
Step 5-1: If mac security prevent opening the file, go to **System Settings**



Step 5-2: On **Privacy & Security** tab, check on these boxex

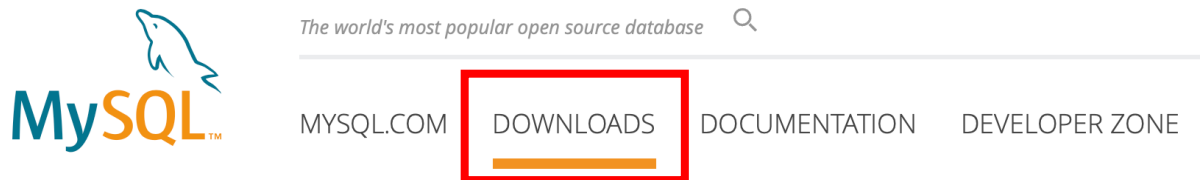


Step 6: Re-open the download file and follow the steps below



Create your password for the "root" user

Installing MySQL on MAC



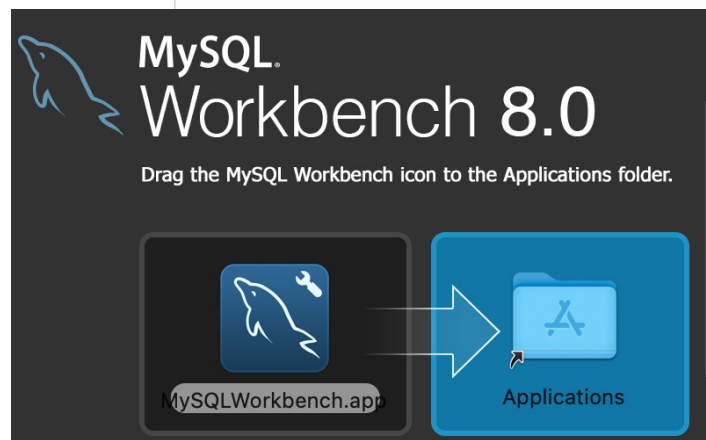
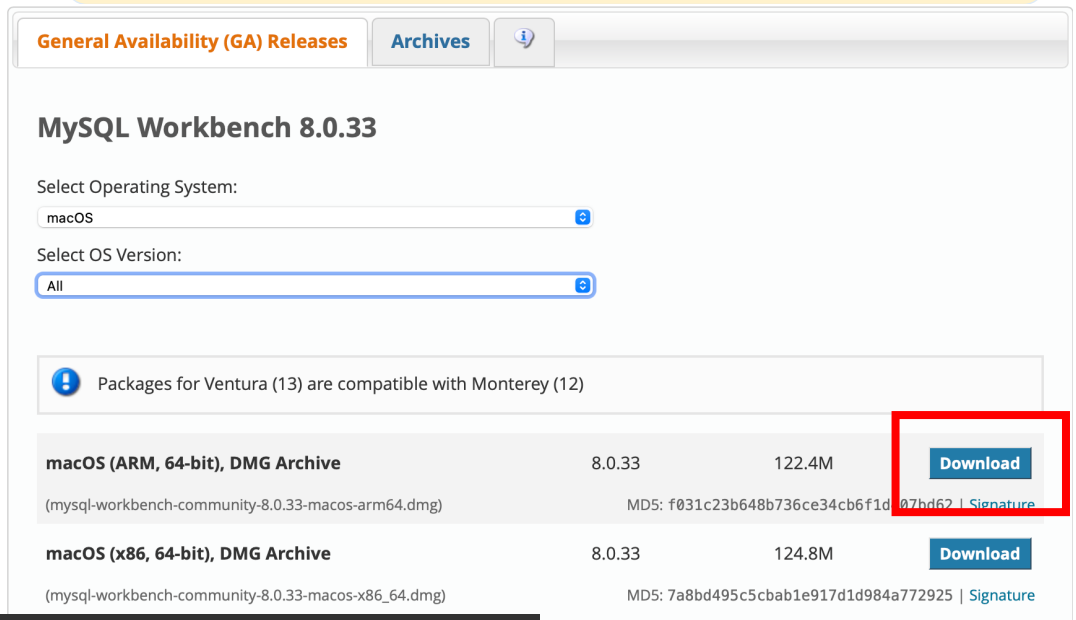
Step 7: Go back to the Download Page

Step 8: Scroll to the bottom of the pages and click [MySQL Community \(GPL\) Downloads »](#)

MySQL Community Downloads

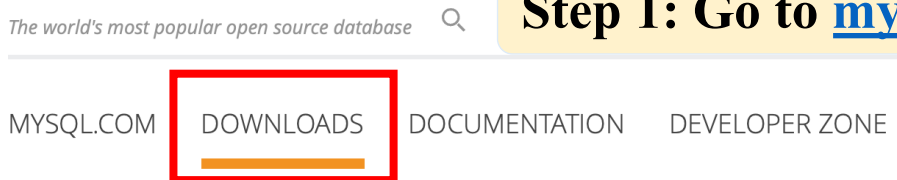
- [MySQL Yum Repository](#)
- [MySQL APT Repository](#)
- [MySQL SUSE Repository](#)
- [MySQL Community Server](#)
- [MySQL Cluster](#)
- [MySQL Router](#)
- [MySQL Shell](#)
- [MySQL Operator](#)
- [MySQL NDB Operator](#)
- [MySQL Workbench](#)
- [MySQL Installer for Windows](#)
- [C API \(libmysqlclient\)](#)
- [Connector/C++](#)
- [Connector/J](#)
- [Connector/NET](#)
- [Connector/Node.js](#)
- [Connector/ODBC](#)
- [Connector/Python](#)
- [MySQL Native Driver for PHP](#)
- [MySQL Benchmark Tool](#)
- [Time zone description tables](#)
- [Download Archives](#)

Step 9: Click on [MySQL Workbench](#) and Download the first link



Step 10: Open the downloaded file and drag to Applications. Then open the app.

Installing MySQL on Windows



Step 1: Go to mysql.com.

Step 2: Click on the **DOWNLOADS** tab

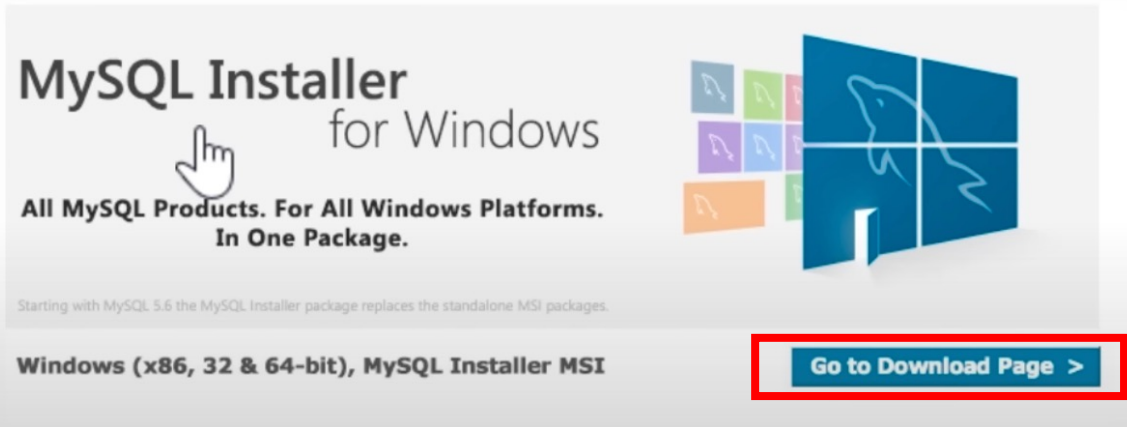
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 - Connector/NET
 - Connector/Node.js
 - Connector/ODBC
 - Connector/Python
 - MySQL Native Driver for PHP
 - MySQL Benchmark Tool
 - Time zone description tables
 - Download Archives

Step 4: Click on **MySQL Community Server** and **Go to Download Page** from the recommended download section

Recommended Download:



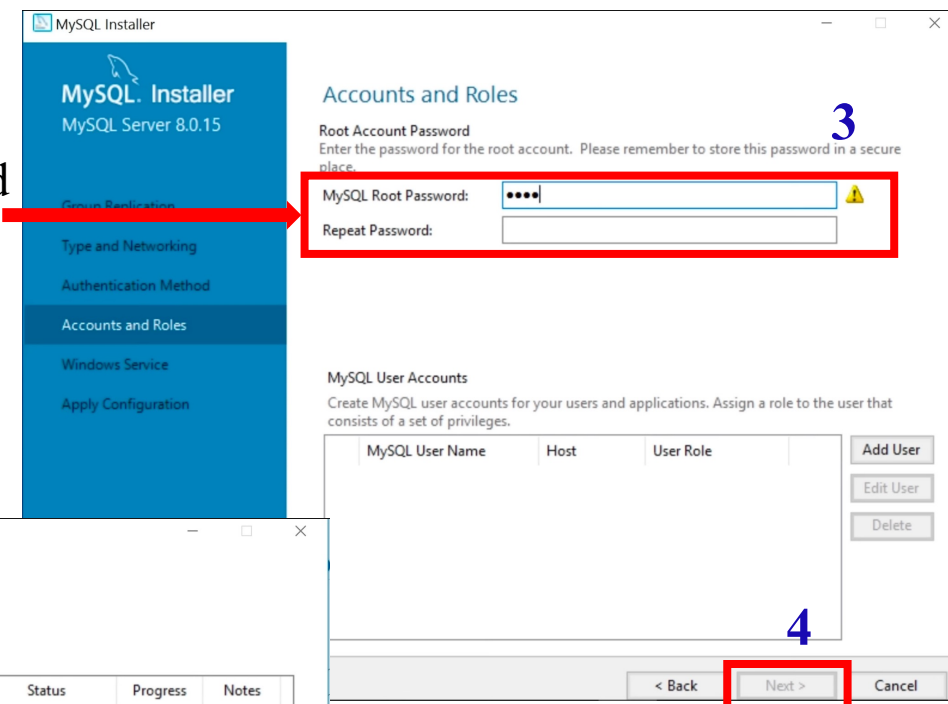
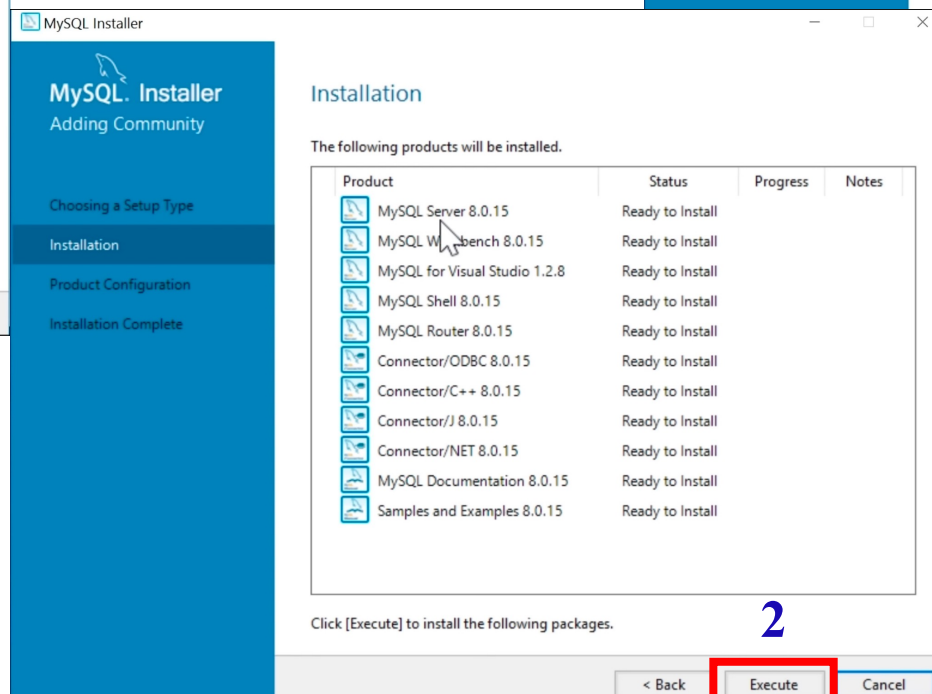
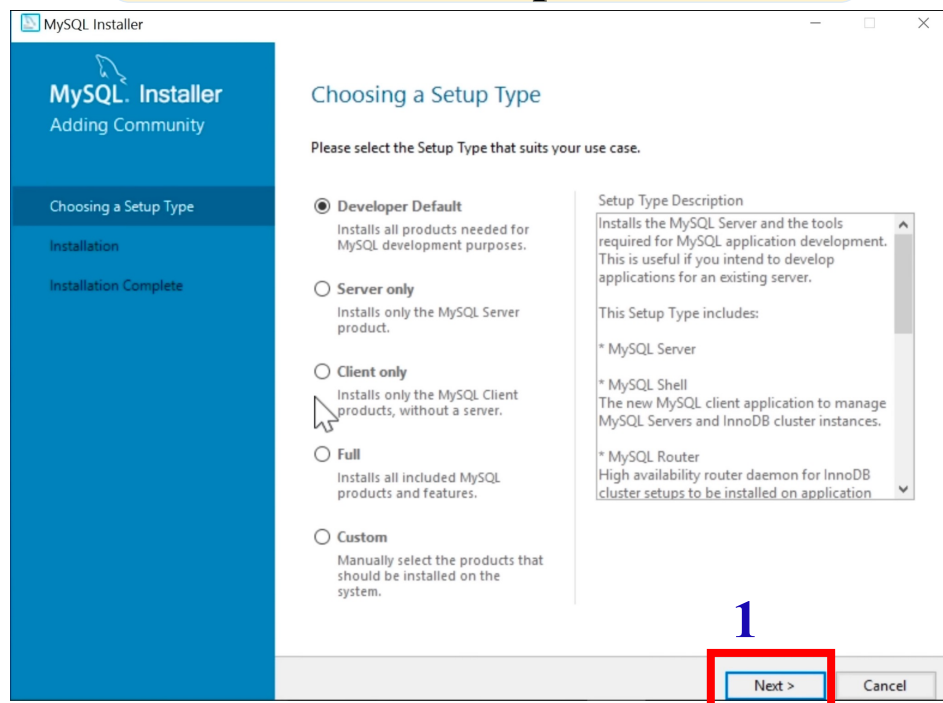
Step 5: Download the first link

Windows (x86, 32-bit), MSI Installer (mysql-installer-web-community-8.0.15.0.msi)	8.0.15	16.4M	Download
		MD5: 66363d23d2474113decf683cf84f5820	
			Signature

Installing MySQL on Windows

Step 5: Open the download file and follow the steps bellow.

Create your password for the “root” user



Installing MySQL on Windows

Step 6: Continue clicking on Next/Execute with default setup

MySQL Installer

Samples and Examples

Connect To Server

Apply Configuration

Connect To Server

Here are the compatible MySQL Svr instances installed in this computer. Please select the ones where the sample schemas and data will be created.

☐ Show MySQL Server instances maybe running in read-only mode

	Server	Port	Arch...	Type	Status
☒	MySQL Server 8.0.15	3306	X64	Stand-alone Server	Connection succeeded

5

Now give us the credentials we should use (needs to have root privileges).
Click OK to continue, or Cancel to abort.

Credentials provided in Server configuration

User name: root

Password:

6

7

Check

All connections succeeded.

Next >

Cancel

Enter your password you set earlier for the “root” user

MySQL Installer

MySQL. Installer
Adding Community

Choosing a Setup Type

Installation

Product Configuration

Installation Complete

Installation Complete

The installation procedure has been completed.

Copy Log to Clipboard

8

☒ Start MySQL Workbench after Setup

☒ Start MySQL Shell after Setup

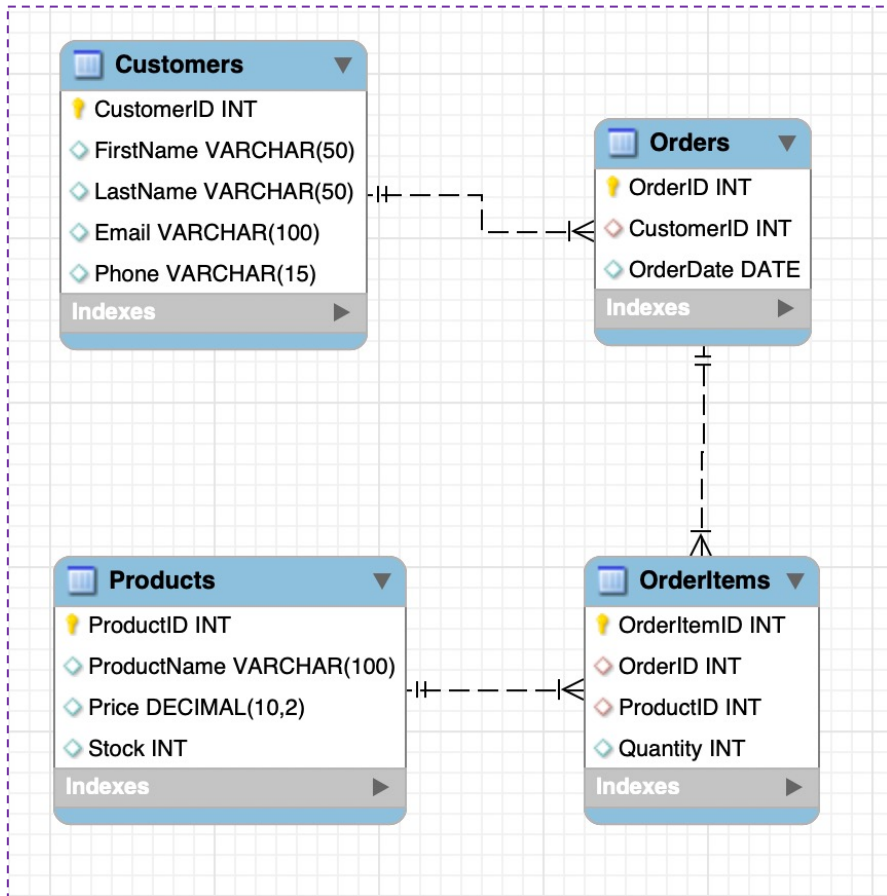
9

Finish

Outline

- 
- **Introduction to Database**
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Create Databases



Storing Data in My SQL Database

Step 1: Design the Database Schema

Step 2: Create Table

Step 3: Insert Data

Step 4: Retrieve Data

Step 5: Update or Delete Data



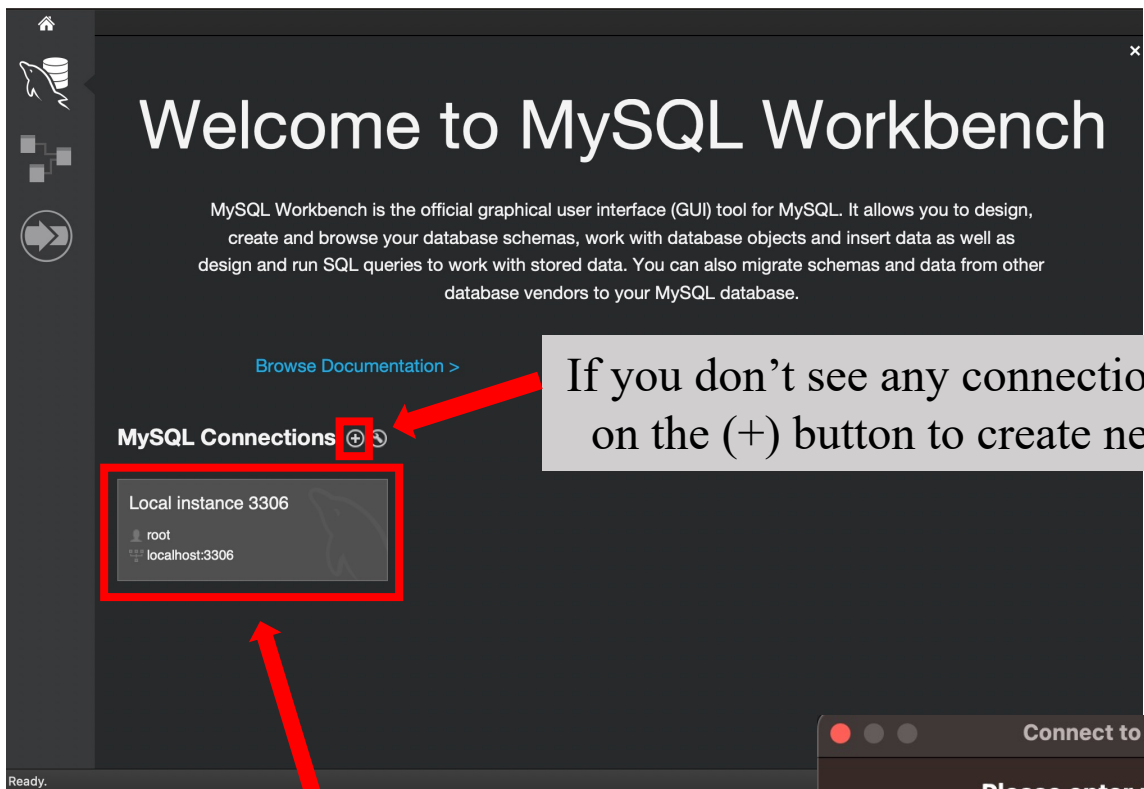
Create Databases

How to store Customer’s Information to Database

CustomerID	FirstName	LastName	Email	Phone
1	John	Doe	john.doe@example.com	555-1234
2	Jane	Smith	jane.smith@example.com	555-5678
3	Alice	Johnson	alice.johnson@example.com	555-8765

Create Databases

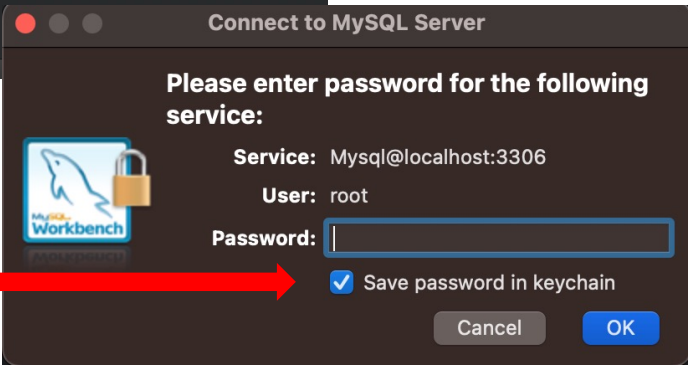
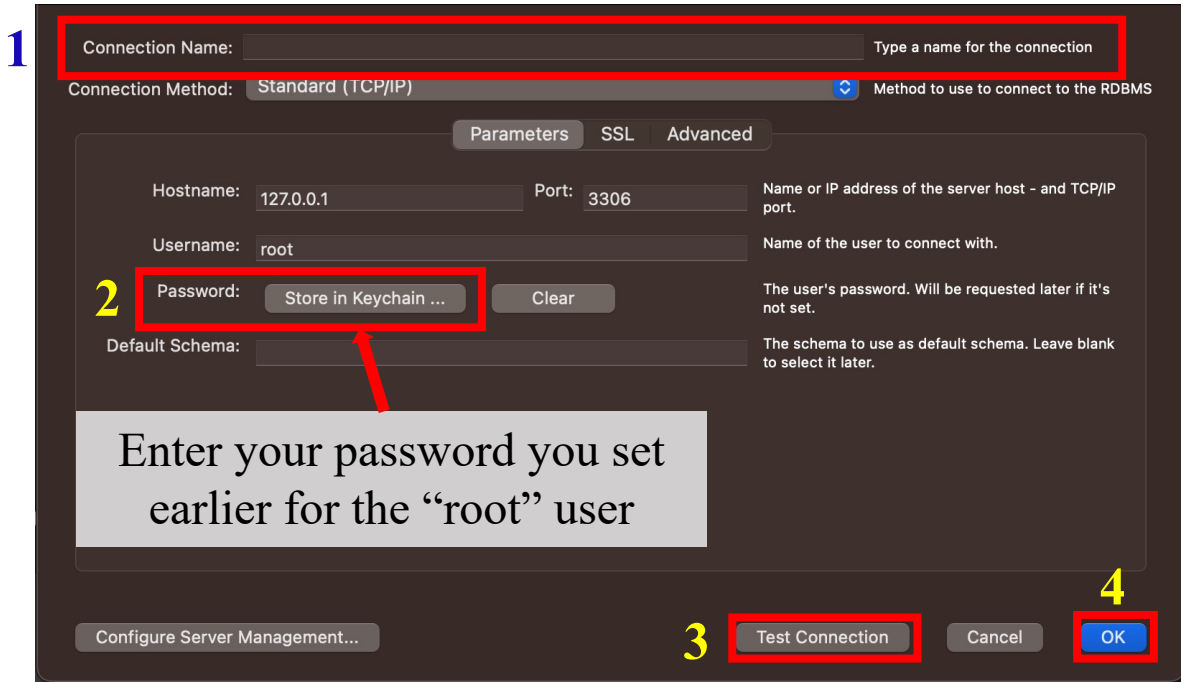
Open MySQL Workbench



If you don't see any connection, click on the (+) button to create new one

You should see the default connection

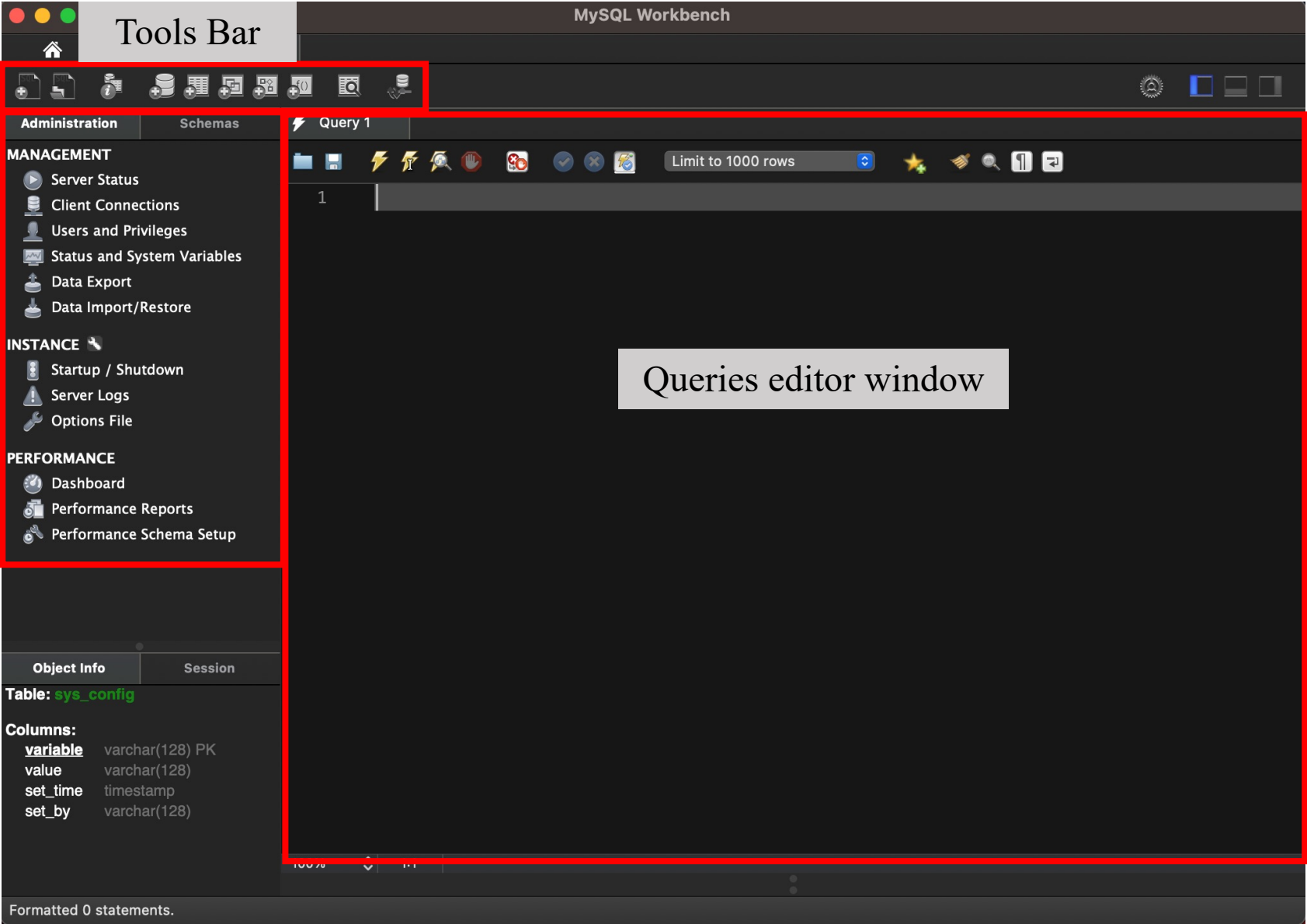
Click and enter your password to access the server



Create Databases

Navigation panels

User Interface



Create a Database

1 Create Database

```
CREATE DATABASE databasename;
```

The **CREATE DATABASE** statement is used to create a new SQL database.

```
DROP DATABASE databasename;
```

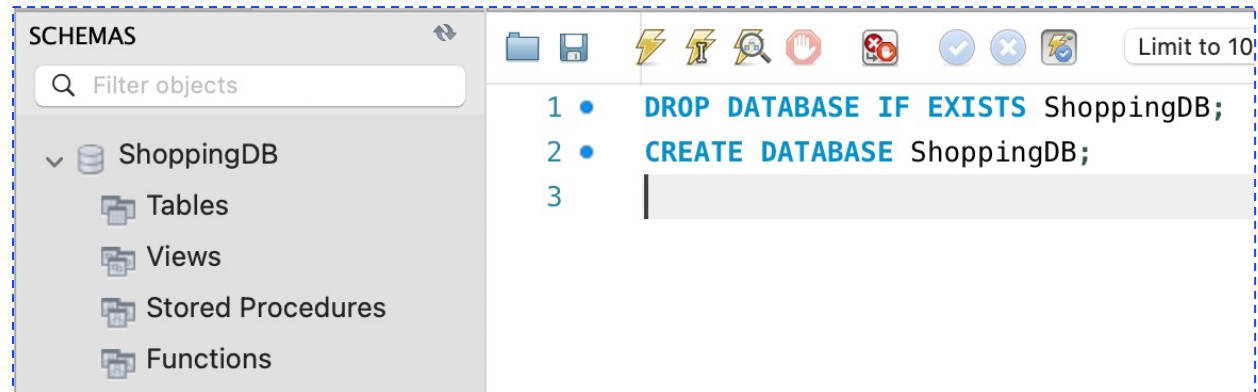
The **DROP DATABASE** statement is used to drop an existing SQL database.

2 Create Table

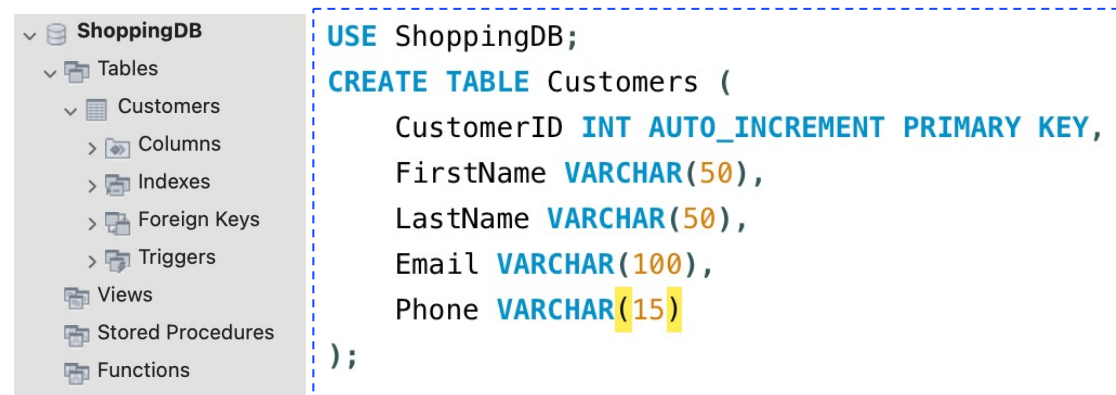
```
CREATE TABLE table_name (
    column1 datatype,
    column2 datatype,
    column3 datatype,
    ....
);
```

The **CREATE TABLE** statement is used to create a new table in a database.

1 Create ShoppingDB Database



2 Create Customers Table in ShoppingDB



Create a Database

3

Insert Data

```
INSERT INTO table_name (column1, column2, column3, ...)
VALUES (value1, value2, value3, ...);
```

4

Query Data

```
SELECT column1, column2, ...
FROM table_name;
```

```
SELECT * FROM table_name;
```

3

Insert data into the Customers Table

```
INSERT INTO Customers (CustomerID, FirstName, LastName, Email, Phone)
VALUES
(1, 'John', 'Doe', 'john.doe@example.com', '555-1234'),
(2, 'Jane', 'Smith', 'jane.smith@example.com', '555-5678'),
(3, 'Alice', 'Johnson', 'alice.johnson@example.com', '555-8765');
```

4

Query Data in the Customers Table

Result Grid

Filter Rows:

Edit:

Export/Import:

	CustomerID	FirstName	LastName	Email	Phone
	1	John	Doe	john.doe@example.com	555-1234
	2	Jane	Smith	jane.smith@example.com	555-5678
	3	Alice	Johnson	alice.johnson@example.com	555-8765

Limit to 10

1 • `SELECT * FROM ShoppingDB.Customers;`

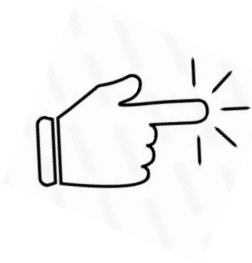


Create Databases

Customer Information

CustomerID	FirstName	LastName	Email	Phone
1	John	Doe	john.doe@example.com	555-1234
2	Jane	Smith	jane.smith@example.com	555-5678
3	Alice	Johnson	alice.johnson@example.com	555-8765

Product Information

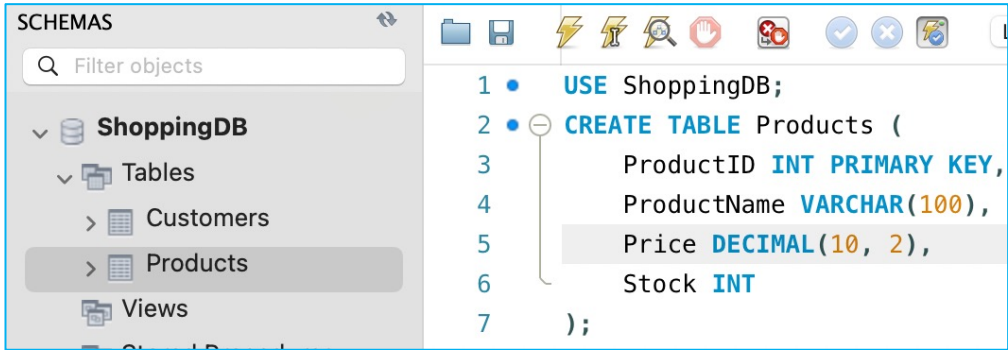


ProductID	ProductName	Price	Stock
1	Laptop	999,99	50
2	Smartphone	499,99	100
3	Tablet	299,99	200

Create a Database

1

Create Products Table in ShoppingDB



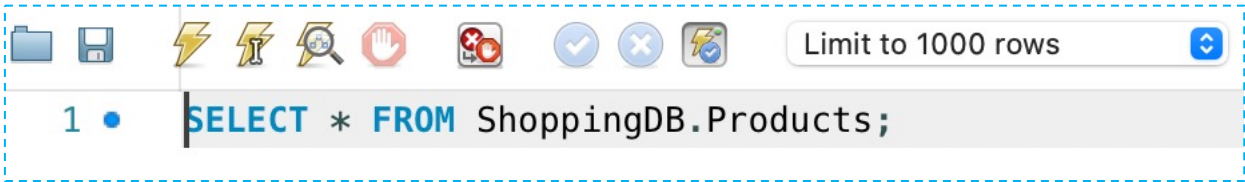
2

Insert data into Product Table

```
1  INSERT INTO Products (ProductID, ProductName, Price, Stock)  
2  VALUES  
3  (1, 'Laptop', 999.99, 50),  
4  (2, 'Smartphone', 499.99, 100),  
5  (3, 'Tablet', 299.99, 200);
```

3

Query Data in the Product Table



Result Grid   Filter Rows: Search				
	ProductID	ProductName	Price	Stock
	1	Laptop	999.99	50
	2	Smartphone	499.99	100
	3	Tablet	299.99	200

Create a Database

Customer

CustomerID	FirstName	LastName	Email	Phone
1	John	Doe	john.doe@example.com	555-1234
2	Jane	Smith	jane.smith@example.com	555-5678
3	Alice	Johnson	alice.johnson@example.com	555-8765

Product

ProductID	ProductName	Price	Stock
1	Laptop	999,99	50
2	Smartphone	499,99	100
3	Tablet	299,99	200

Customer might buy many products. How?

OrderID	CustomerName	OrderDate	ProductName	Quantity
1	John Doe	2023-07-01	Laptop	1
1	John Doe	2023-07-01	Tablet	2
2	Jane Smith	2023-07-02	Smartphone	1

Create Databases

Customer

CustomerID	FirstName	LastName	Email	Phone
1	John	Doe	john.doe@example.com	555-1234
2	Jane	Smith	jane.smith@example.com	555-5678
3	Alice	Johnson	alice.johnson@example.com	555-8765

Order

OrderID	CustomerID	OrderDate
1	1	2023-07-01
2	2	2023-07-02

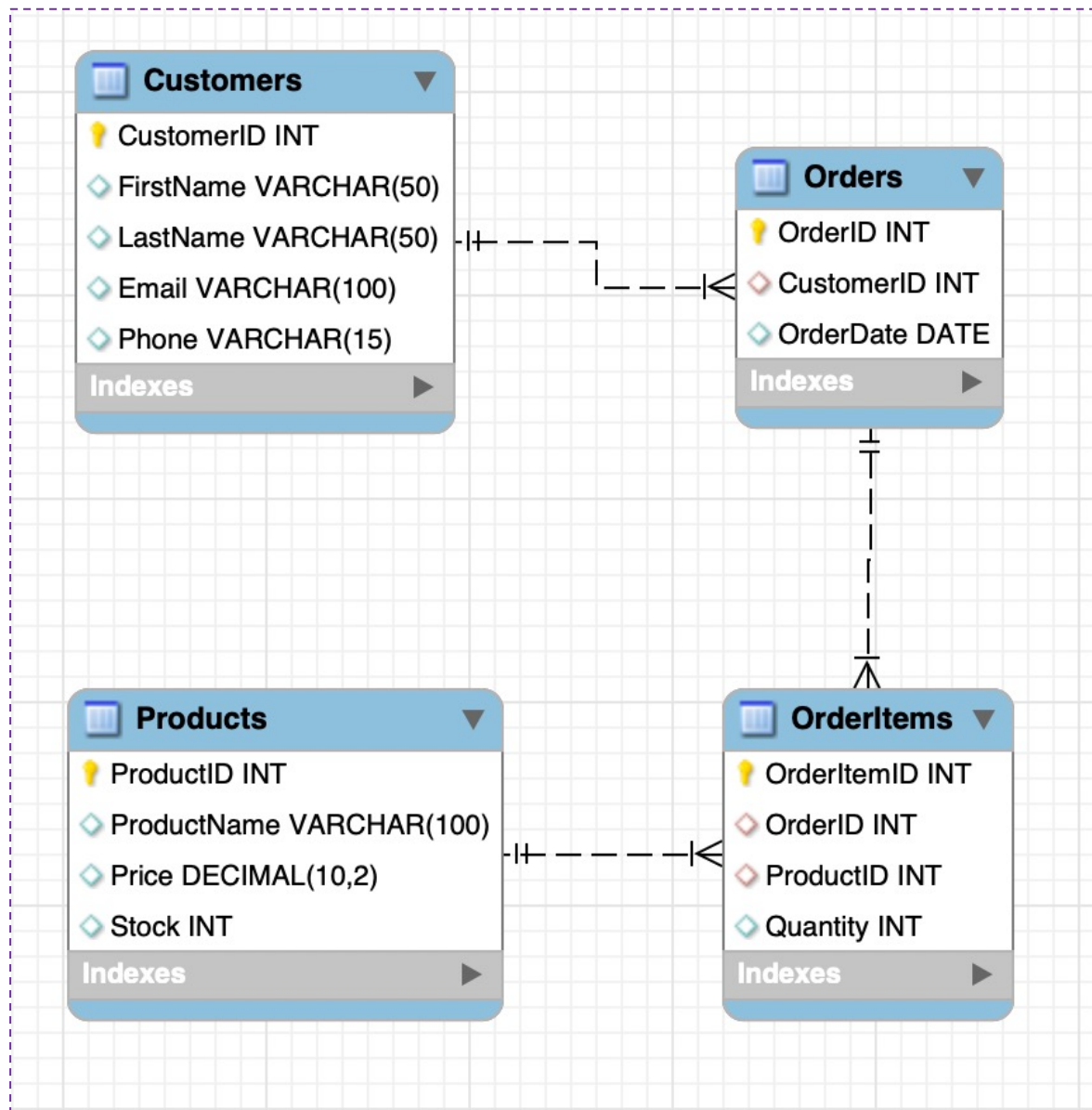
Product

ProductID	ProductName	Price	Stock
1	Laptop	999,99	50
2	Smartphone	499,99	100
3	Tablet	299,99	200

OrderItem

OrderItemID	OrderID	ProductID	Quantity
1	1	1	1
2	1	3	2
3	2	2	1

Create Databases



Create a Database

1 Create Order Table in ShoppingDB

```
1  USE ShoppingDB;
2  • CREATE TABLE Orders (
3      OrderID INT PRIMARY KEY,
4      CustomerID INT,
5      OrderDate DATE,
6      FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID)
7  );
```

2 Create OrderItems Table

```
9  • USE ShoppingDB;
10 • CREATE TABLE OrderItems (
11     OrderItemID INT PRIMARY KEY,
12     OrderID INT,
13     ProductID INT,
14     Quantity INT,
15     FOREIGN KEY (OrderID) REFERENCES Orders(OrderID),
16     FOREIGN KEY (ProductID) REFERENCES Products(ProductID)
17 );
```

3 Insert Data

```
1  • INSERT INTO Customers (CustomerID, FirstName, LastName, Email, Phone)
2  VALUES
3      (1, 'John', 'Doe', 'john.doe@example.com', '555-1234'),
4      (2, 'Jane', 'Smith', 'jane.smith@example.com', '555-5678'),
5      (3, 'Alice', 'Johnson', 'alice.johnson@example.com', '555-8765');
6
7  • INSERT INTO Products (ProductID, ProductName, Price, Stock)
8  VALUES
9      (1, 'Laptop', 999.99, 50),
10     (2, 'Smartphone', 499.99, 100),
11     (3, 'Tablet', 299.99, 200);
12
13 • INSERT INTO Orders (OrderID, CustomerID, OrderDate)
14 VALUES
15     (1, 1, '2023-07-01'),
16     (2, 2, '2023-07-02');
17
18 • INSERT INTO OrderItems (OrderItemID, OrderID, ProductID, Quantity)
19 VALUES
20     (1, 1, 1, 1),
21     (2, 1, 3, 2),
22     (3, 2, 2, 1);
```

Entity Relationship Diagram

Database Server Tools Scripting

1

Connect to Database...

Manage Connections...

Reverse Engineer... R

Schema Transfer Wizard...

Migration Wizard...

Edit Type Mappings for Generic Migration...

Search Table Data...

Reverse Engineer Database

2

Set Parameters for Connecting to a DBMS

Connection Options

Connect to DBMS

Select Schemas

Retrieve Objects

Select Objects

Reverse Engineer

Results

Stored Connection: Local instance 3306

Connection Method: Standard (TCP/IP)

Parameters SSL Advanced

Hostname: localhost

Port: 3306

Name or IP address of the server host - and TCP/IP port.

Username: root

Name of the user to connect with.

Password: Store in Keychain ...

Clear

The user's password. Will be requested later if it's not set.

Reverse Engineer Database

3

Connect to DBMS and Fetch Information

Connection Options

Connect to DBMS

Select Schemas

Retrieve Objects

Select Objects

Reverse Engineer

Results

The following tasks will now be executed. Please monitor the execution. Press Show Logs to see the execution logs.

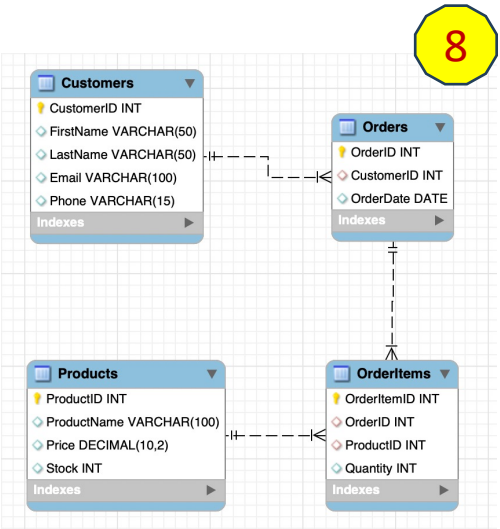
Connect to DBMS

Retrieve Schema List from Database

Check Common Server Configuration Issues

Execution Completed Successfully

Fetch finished.



Reverse Engineer Database

4

Select Schemas to Reverse Engineer

Connection Options

Connect to DBMS

Select Schemas

Retrieve Objects

Select Objects

Reverse Engineer

Results

Select the schemas you want to include:

ShoppingDB

Reverse Engineer Database

5

Retrieve and Reverse Engineer Schema Objects

Connection Options

Connect to DBMS

Select Schemas

Retrieve Objects

Select Objects

Reverse Engineer

Results

The following tasks will now be executed. Please monitor the execution. Press Show Logs to see the execution logs.

Retrieve Objects from Selected Schemas

Check Results

Retrieval Completed Successfully

Finished.

Reverse Engineer Database

6

Select Objects to Reverse Engineer

Connection Options

Connect to DBMS

Select Schemas

Retrieve Objects

Select Objects

Reverse Engineer

Results

Import MySQL Table Objects

4 Total Objects, 4 Selected

Reverse Engineer Database

7

Reverse Engineering Progress

Connection Options

Connect to DBMS

Select Schemas

Retrieve Objects

Select Objects

Reverse Engineer

Results

The following tasks will now be executed. Please monitor the execution. Press Show Logs to see the execution logs.

Reverse Engineer Selected Objects

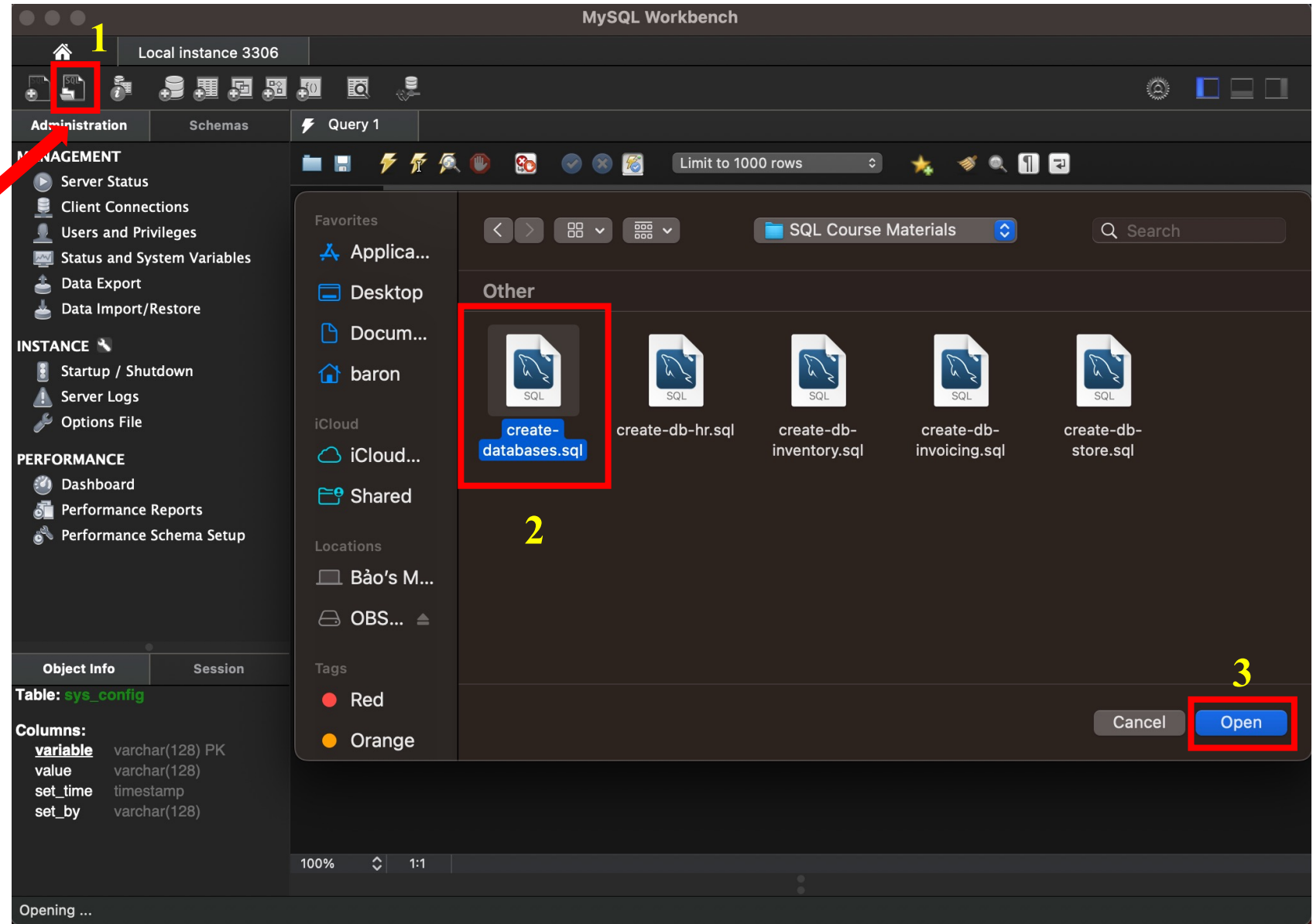
Place Objects on Diagram

Operation Completed Successfully

Create Databases From Script

Download the pre-created database script here

Click to open the downloaded script



Create Databases

Click the thunder icon to execute and create our databases

NOTE: This button can be used to execute **SELECTED** SQL code. To execute **ALL** code, **UNSELECT** everything

Go to Schemas tab on the Navigation panels and click the refresh icon to see the created schemas

MySQL Workbench

Local instance 3306

Administration Schemas Query 1 create-databases

Limit to 1000 rows

SCHEMAS

sql_hr

sql_inventory

sql_invoicing

sql_store

sys

1 DROP DATABASE IF EXISTS `sql\_invoicing`;

2 CREATE DATABASE `sql\_invoicing`;

3 USE `sql\_invoicing`;

4

5 SET NAMES utf8 ;

6 SET character_set_client = utf8mb4 ;

7

8 CREATE TABLE `payment\_methods` (

9 `payment\_method\_id` tinyint(4) NOT NULL AUTO_INCREMENT,

10 `name` varchar(50) NOT NULL,

11 PRIMARY KEY (`payment\_method\_id`)

12) ENGINE=InnoDB AUTO_INCREMENT=5 DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;

13 INSERT INTO `payment\_methods` VALUES (1,'Credit Card');

14 INSERT INTO `payment\_methods` VALUES (2,'Cash');

15 INSERT INTO `payment\_methods` VALUES (3,'PayPal');

16 INSERT INTO `payment\_methods` VALUES (4,'Wire Transfer');

17

18 CREATE TABLE `clients` (

19 `client\_id` int(11) NOT NULL,

20 `name` varchar(50) NOT NULL,

21 `address` varchar(50) NOT NULL,

22 `city` varchar(50) NOT NULL,

Object Info

Session

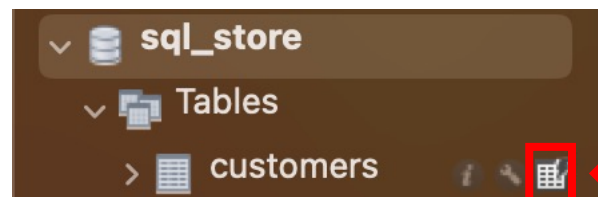
No object selected

Action Output

	Time	Action	Response	Duration / Fetch Time
152	00:57:56	INSERT INTO `products` VALUES (1,'Foam Dinner Plate',70,1.21)	1 row(s) affected	0.00055 sec
153	00:57:56	INSERT INTO `products` VALUES (2,'Pork - Bacon,back Peameal',49,4.65)	1 row(s) affected	0.00028 sec
154	00:57:56	INSERT INTO `products` VALUES (3,'Lettuce - Romaine, Heart',38,3.35)	1 row(s) affected	0.00020 sec
155	00:57:56	INSERT INTO `products` VALUES (4,'Brocolinni - Gaylan, Chinese',90,4.53)	1 row(s) affected	0.00025 sec
156	00:57:56	INSERT INTO `products` VALUES (5,'Sauce - Ranch Dressing',94,1.63)	1 row(s) affected	0.00026 sec
157	00:57:56	INSERT INTO `products` VALUES (6,'Petit Baguette',14,2.39)	1 row(s) affected	0.00028 sec
158	00:57:56	INSERT INTO `products` VALUES (7,'Sweet Pea Sprouts',98,3.29)	1 row(s) affected	0.00024 sec
159	00:57:56	INSERT INTO `products` VALUES (8,'Island Oasis - Raspberry',26,0.74)	1 row(s) affected	0.00038 sec
160	00:57:56	INSERT INTO `products` VALUES (9,'Longan',67,2.26)	1 row(s) affected	0.00024 sec
161	00:57:56	INSERT INTO `products` VALUES (10,'Broom - Push',6,1.09)	1 row(s) affected	0.00025 sec

Active schema was cleared

Explore the data



Go to Schemas tab on the Navigation panel, select any schemas and table and click on this icon to view the data.

[illegible]

Outline

- **Introduction to Database**
- **How to install MySQL**
- **How to create Database**
- **How to use SQL for data analysis**
- **Kahoot Quiz**
- **Summary**





SQL Queries

SELECT

SQL (Structured Query Language) is a powerful tool for data analysis. It allows you to query and manipulate databases efficiently



SELECT

1/ Select all data from a table

```
SELECT * FROM <schema>.<table>;
```

```
SELECT * FROM sql_store.orders;
```

NOTE: You can use the keyword **USE** to specify the schema you want to query from beforehand

```
USE sql_store;
SELECT * FROM orders;
```

order_id	customer_id	order_date	status	comments	shipped_date	shipper_id	
1	6	2019-01-30	1	NULL	NULL	NULL	
2	7	2018-08-02	2	NULL	2018-08-03	4	
3	8	2017-12-01	1	NULL	NULL	NULL	
4	2	2017-01-22	1	NULL	NULL	NULL	
5	5	2017-08-25	2		2017-08-26	3	
6	10	2018-11-18	1	Aliquam erat volutpat. In congue.	NULL	NULL	
7	2	2018-09-22	2	NULL	2018-09-23	4	
8	5	2018-06-08	1	Mauris enim leo, rhoncus sed, vestibulum sit am...	NULL	NULL	
9	10	2017-07-05	2	Nulla mollis molestie lorem. Quisque ut erat.	2017-07-06	1	
10	6	2018-04-22	2	NULL	2018-04-23	2	
NULL	NULL	NULL	NULL	NULL	NULL	NULL	

2/ Select specific columns

```
SELECT <column>,<column>,<...>
FROM <table>;
```

```
SELECT first_name, last_name
FROM customers;
```

first_name	last_name	
Babara	MacCaffrey	
Ines	Brushfield	
Freddi	Boagey	
Ambur	Roseburgh	
Clemmie	Betchley	
Elka	Twiddell	
Ilene	Dowson	
Thacher	Naseby	
Romola	Rumgay	
Levy	Mynett	

SELECT

3/ Select with modify values

```
SELECT
  points,
  points + 10,
  points / 10
FROM customers;
```

	points	points + 10	points / 10	
	2273	2283	227.3000	
	947	957	94.7000	
	2967	2977	296.7000	
	457	467	45.7000	
	3675	3685	367.5000	
	3073	3083	307.3000	
	1672	1682	167.2000	
	205	215	20.5000	
	1486	1496	148.6000	
	796	806	79.6000	

4/ Select with Alias

```
SELECT <column> AS <alias>
FROM <table>;
```

```
SELECT
  points,
  points * 110 / 100,
  points * 110 / 100 AS VAT
FROM customers;
```

	points	points * 110 / 100	VAT	
	2273	2500.3000	2500.3000	
	947	1041.7000	1041.7000	
	2967	3263.7000	3263.7000	
	457	502.7000	502.7000	
	3675	4042.5000	4042.5000	
	3073	3380.3000	3380.3000	
	1672	1839.2000	1839.2000	
	205	225.5000	225.5000	
	1486	1634.6000	1634.6000	
	796	875.6000	875.6000	



SELECT

5/ Select unique values of a column

```
SELECT state
FROM customers;
```

state
VA
VA
CO
FL
TX
IL
TN
FL
CA
GA

```
SELECT DISTINCT state
FROM customers;
```

state
VA
CO
FL
TX
IL
TN
CA
GA

6/ Select all the products:

- Name
- Unit price
- New price (Unit price * 1.1)

```
SELECT
name,
unit_price,
unit_price * 1.1 AS new_price
FROM products;
```

name	unit_price	new_price
Foam Dinner Plate	1.21	1.331
Pork - Bacon,back Peameal	4.65	5.115
Lettuce - Romaine, Heart	3.35	3.685
Brocolinni - Gaylan, Chinese	4.53	4.983
Sauce - Ranch Dressing	1.63	1.793
Petit Baguette	2.39	2.629
Sweet Pea Sprouts	3.29	3.619
Island Oasis - Raspberry	0.74	0.814
Longan	2.26	2.486
Broom - Push	1.09	1.199



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SQL Queries

WHERE

1/ Select data with condition

```
SELECT <column>
FROM <table>
WHERE <condition>;
```

```
SELECT *  
FROM customers  
WHERE points > 3000;
```

[illegible]

2/ Comparison Operators

<, >, <=, >=, =, !=, <>

3/ Select every customer from state 'VA'

```
SELECT *  
FROM customers  
WHERE state = 'VA';
```

[illegible]

4/ Select every customer not from state 'VA'

```
SELECT *
FROM customers
WHERE state != 'VA';
```

[illegible]



WHERE

5/ Get the orders placed after 2018

```
SELECT *  
FROM orders  
WHERE order_date >= '2018-01-01';
```

order_id	customer_id	order_date	status	comments	shipped_date	shipper_id
1	6	2019-01-30	1	NULL	NULL	NULL
2	7	2018-08-02	2	NULL	2018-08-03	4
6	10	2018-11-18	1	Aliquam erat volutpat. In congue.	NULL	NULL
7	2	2018-09-22	2	NULL	2018-09-23	4
8	5	2018-06-08	1	Mauris enim leo, rhoncus sed, vestibulum sit am...	NULL	NULL
10	6	2018-04-22	2	NULL	2018-04-23	2
NULL	NULL	NULL	NULL	NULL	NULL	NULL

7/ From order_items table, get order_id #6 and total price is less than 30

```
SELECT *, unit_price * quantity AS total_price  
FROM order_items  
WHERE  
    order_id = 6 AND unit_price * quantity < 30;
```

order_id	product_id	quantity	unit_price	total_price
6	2	4	3.28	13.12
6	3	4	7.46	29.84
6	5	1	3.45	3.45

6/ The AND, OR, and NOT Operators

AND Truth Table

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

OR Truth Table

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

NOT Truth Table

A	B
0	1
1	0

NOT '<' is '>='
NOT '>' is '<='
NOT 'AND' is 'OR'
NOT 'OR' is 'AND'
NOT '=' is '!='
NOT '<>' is '='



SQL Queries

IN - BETWEEN

IN - BETWEEN

1/ Select all customer from state (use OR): VA, GA, FL

```
SELECT *
FROM customers
WHERE
    state = 'VA' OR state = 'GA' OR state = 'FL';
```

[illegible]

2/ Select all customer from state (use IN): VA, GA, FL

```
SELECT *
FROM customers
WHERE
    state IN ('VA', 'GA', 'FL');
```

[illegible]



IN - BETWEEN

3/ Select all customer from state VA, GA, FL
points greater than 300 but less than 2000

```
SELECT * FROM customers
WHERE
    state IN ('VA', 'GA', 'FL')
    AND points > 300
    AND points < 2000;
```

customer_id	first_name	last_name	birth_date	phone	address	city	state	points
2	Ines	Brushfield	1986-04-13	804-427-9456	14187 Commercial Trail	Hampton	VA	947
4	Ambur	Roseburgh	1974-04-14	407-231-8017	30 Arapahoe Terrace	Orlando	FL	457
10	Levy	Mynett	1969-10-13	404-246-3370	68 Lawn Avenue	Atlanta	GA	796

4/ Select all customer from state VA, GA, FL
points BETWEEN 300 - 2000

```
SELECT * FROM customers
WHERE
    state IN ('VA', 'GA', 'FL')
    AND points BETWEEN 300 AND 2000;
```

customer_id	first_name	last_name	birth_date	phone	address	city	state	points
2	Ines	Brushfield	1986-04-13	804-427-9456	14187 Commercial Trail	Hampton	VA	947
4	Ambur	Roseburgh	1974-04-14	407-231-8017	30 Arapahoe Terrace	Orlando	FL	457
10	Levy	Mynett	1969-10-13	404-246-3370	68 Lawn Avenue	Atlanta	GA	796



SQL Queries

IS NULL - ORDER BY - LIMIT

IS NULL - ORDER BY - LIMIT

1/ Select all customer don't have a phone number

```
SELECT *
FROM customers
WHERE phone IS NULL;
```

[illegible]

2/ Select all customer who have a phone number

```
SELECT *
FROM customers
WHERE phone IS NOT NULL;
```

	customer_id	first_name	last_name	birth_date	phone	address	city	state	points
1		Babara	MacCaffrey	1986-03-28	781-932-9754	0 Sage Terrace	Waltham	VA	2273
2		Ines	Brushfield	1986-04-13	804-427-9456	14187 Commercial Trail	Hampton	VA	947
3		Freddi	Boagey	1985-02-07	719-724-7869	251 Springs Junction	Colorado Springs	CO	2967
4		Ambur	Roseburgh	1974-04-14	407-231-8017	30 Arapahoe Terrace	Orlando	FL	457
6		Elka	Twiddell	1991-09-04	312-480-8498	7 Manley Drive	Chicago	IL	3073
7		Ilene	Dowson	1964-08-30	615-641-4759	50 Lillian Crossing	Nashville	TN	1672
8		Thacher	Naseby	1993-07-17	941-527-3977	538 Mosinee Center	Sarasota	FL	205
9		Romola	Rumgay	1992-05-23	559-181-3744	3520 Ohio Trail	Visalia	CA	1486
10		Levy	Mynett	1969-10-13	404-246-3370	68 Lawn Avenue	Atlanta	GA	796
	SALES 1	SALES 2	SALES 3	SALES 4	SALES 5	SALES 6	SALES 7	SALES 8	

IS NULL - ORDER BY - LIMIT

3/ Order the customer by points in ascending order

```
SELECT * FROM customers
ORDER BY points;
```

customer_id	first_name	last_name	birth_date	phone	address	city	state	points
8	Thacher	Naseby	1993-07-17	941-527-3977	538 Mosinee Center	Sarasota	FL	205
4	Ambur	Roseburgh	1974-04-14	407-231-8017	30 Arapahoe Terrace	Orlando	FL	457
10	Levy	Mynett	1969-10-13	404-246-3370	68 Lawn Avenue	Atlanta	GA	796
2	Ines	Brushfield	1986-04-13	804-427-9456	14187 Commercial Trail	Hampton	VA	947
9	Romola	Rumgay	1992-05-23	559-181-3744	3520 Ohio Trail	Visalia	CA	1486
7	Ilene	Dowson	1964-08-30	615-641-4759	50 Lillian Crossing	Nashville	TN	1672
1	Babara	MacCaffrey	1986-03-28	781-932-9754	0 Sage Terrace	Waltham	VA	2273
3	Freddi	Boagey	1985-02-07	719-724-7869	251 Springs Junction	Colorado Springs	CO	2967
6	Elka	Twiddell	1991-09-04	312-480-8498	7 Manley Drive	Chicago	IL	3073
5	Clemmie	Betchley	1973-11-07	NULL	5 Spohn Circle	Arlington	TX	3675

4/ Get the customer have points < 1000 and order them in descending order by points

```
SELECT * FROM customers
WHERE points < 1000
ORDER BY points DESC;
```

customer_id	first_name	last_name	birth_date	phone	address	city	state	points
2	Ines	Brushfield	1986-04-13	804-427-9456	14187 Commercial Trail	Hampton	VA	947
10	Levy	Mynett	1969-10-13	404-246-3370	68 Lawn Avenue	Atlanta	GA	796
4	Ambur	Roseburgh	1974-04-14	407-231-8017	30 Arapahoe Terrace	Orlando	FL	457
8	Thacher	Naseby	1993-07-17	941-527-3977	538 Mosinee Center	Sarasota	FL	205

IS NULL - ORDER BY - LIMIT

5/ Get three customer with the highest points

```
SELECT * FROM customers
ORDER BY points DESC
LIMIT 3;
```

customer_id	first_name	last_name	birth_date	phone	address	city	state	points
5	Clemmie	Betchley	1973-11-07	NULL	5 Spohn Circle	Arlington	TX	3675
6	Elka	Twiddell	1991-09-04	312-480-8498	7 Manley Drive	Chicago	IL	3073
3	Freddi	Boagey	1985-02-07	719-724-7869	251 Springs Junction	Colorado Springs	CO	2967

6/ Get four customer with the highest points (not including the first 3)

```
SELECT * FROM customers
ORDER BY points DESC
LIMIT 3, 4;
```

customer_id	first_name	last_name	birth_date	phone	address	city	state	points
1	Babara	MacCaffrey	1986-03-28	781-932-9754	0 Sage Terrace	Waltham	VA	2273
7	Ilene	Dowson	1964-08-30	615-641-4759	50 Lillian Crossing	Nashville	TN	1672
9	Romola	Rumgay	1992-05-23	559-181-3744	3520 Ohio Trail	Visalia	CA	1486
2	Ines	Brushfield	1986-04-13	804-427-9456	14187 Commercial Trail	Hampton	VA	947



SQL Queries

LIKE - REGEXP

1/ Select all customer whose last name start with 'B'

```
SELECT *
FROM customers
WHERE last_name LIKE 'B%';
```

[illegible]

NOTE: ‘%’ is used to represent any number of characters

2/ Select all customer who have a 'b' in their last name

```
SELECT *
FROM customers
WHERE last_name LIKE '%B%';
```

customer_id	first_name	last_name	birth_date	phone	address	city	state	points
2	Ines	Brushfield	1986-04-13	804-427-9456	14187 Commercial Trail	Hampton	VA	947
3	Frank	Boagey	1985-02-07	719-724-7869	251 Springs Junction	Colorado Springs	CO	2967
4	Anna	Roseburgh	1974-04-14	407-231-8017	30 Arapahoe Terrace	Orlando	FL	457
5	Clara	Betchley	1973-11-07	NULL	5 Spohn Circle	Arlington	TX	3675
8	Thomas	Naseby	1993-07-17	941-527-3977	538 Mosinee Center	Sarasota	FL	205



3/ Select all customers whose last name is 6 letters long and end with ‘y’

```
SELECT *  
FROM customers  
WHERE last_name LIKE '____y';
```

customer_id	first_name	last_name	birth_date	phone	address	city	state	points
3	Frank	Boagey	1985-02-07	719-724-7869	251 Springs Junction	Colorado Springs	CO	2967
8	Thomas	Naseby	1993-07-17	941-527-3977	538 Mosinee Center	Sarasota	FL	205
9	Rumgay	Rumgay	1992-05-23	559-181-3744	3520 Ohio Trail	Visalia	CA	1486

NOTE: ‘_’ is used to represent one characters

4/ Get the customer whose
-- addresses contain ‘TRAIL’ or ‘AVENUE’
-- phone numbers end with 9---

```
SELECT *  
FROM customers  
WHERE  
    address LIKE '%trail%' OR  
    address LIKE '%avenue%';
```

customer_id	first_name	last_name	birth_date	phone	address	city	state	points
2	Ines	Brushfield	1986-04-13	804-427-9456	14187 Commercial Trail	Hampton	VA	947
9	Rumgay	Rumgay	1992-05-23	559-181-3744	3520 Ohio Trail	Visalia	CA	1486
10	Levy	Mynett	1969-10-13	404-246-3370	68 Lawn Avenue	Atlanta	GA	796

```
SELECT *  
FROM customers  
WHERE phone LIKE '%9____';
```

customer_id	first_name	last_name	birth_date	phone	address	city	state	points
1	Baer	MacCaffrey	1986-03-28	781-932-9754	0 Sage Terrace	Waltham	VA	2273
2	Ines	Brushfield	1986-04-13	804-427-9456	14187 Commercial Trail	Hampton	VA	947



REGEXP

1/ Get the customer whose first name are ELKA or AMBUR

```
SELECT *  
FROM customers  
WHERE first_name REGEXP 'ELKA | AMBUR';
```

customer_id	first_name	last_name	birth_date	phone	address	city	state	points
4	Ambur	Roseburgh	1974-04-14	407-231-8017	30 Arapahoe Terrace	Orlando	FL	457
6	Elka	Twiddell	1991-09-04	312-480-8498	7 Manley Drive	Chicago	IL	3073

NOTE: ‘|’ is used to represent alternation.
‘|’ = OR

2/ Get the customer whose last name end with EY or ON

```
SELECT *  
FROM customers  
WHERE last_name REGEXP 'EY$|ON$';
```

customer_id	first_name	last_name	birth_date	phone	address	city	state	points
1	Babara	MacCaffrey	1986-03-28	781-932-9754	0 Sage Terrace	Waltham	VA	2273
3	Freddi	Boagey	1985-02-07	719-724-7869	251 Springs Junction	Colorado Springs	CO	2967
5	Clemmie	Betchley	1973-11-07	NULL	5 Spohn Circle	Arlington	TX	3675
7	Ilene	Dowson	1964-08-30	615-641-4759	50 Lillian Crossing	Nashville	TN	1672

NOTE: ‘\$’ is used to match the end of a string.

3/ Get the customer whose last name start with MY or contains SE

```
SELECT *
FROM customers
WHERE last_name REGEXP '^MY|SE';
```

customer_id	first_name	last_name	birth_date	phone	address	city	state	points
4	Ambur	Roseburgh	1974-04-14	407-231-8017	30 Arapahoe Terrace	Orlando	FL	457
8	Thacher	Naseby	1993-07-17	941-527-3977	538 Mosinee Center	Sarasota	FL	205
10	Levy	Mynett	1969-10-13	404-246-3370	68 Lawn Avenue	Atlanta	GA	796
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

NOTE: '^' is used to match the beginning of a string.

4/ Get the customer whose last name contain B followed by R or U

```
SELECT *
FROM customers
WHERE last_name REGEXP 'B[RU]';
```

customer_id	first_name	last_name	birth_date	phone	address	city	state	points
2	Ines	Brushfield	1986-04-13	804-427-9456	14187 Commercial Trail	Hampton	VA	947
4	Ambur	Roseburgh	1974-04-14	407-231-8017	30 Arapahoe Terrace	Orlando	FL	457
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

NOTE: '[' is used to match any characters in it.

REGEX basic syntaxs

REGEX SYNTAX	MEANING	EXAMPLE	MATCHES	DOES NOT MATCH
.	Any single character	go.gle	google, goggle	gogle
[abc]	Any of these character	analy[zs]e	analyse, analyze	analyxe
[a-z]	Any character in this range	demo[2-4]	demo2, demo3	demo1, demo5
[^abc]	None of these characters	analy[^zs]e	analyxe	analyse, analyze
[^a-z]	Not a character in this range	demo[^2-4]	demo1, demo5	demo2, demo3
	Or	demo example	demo, demos, example	test
^	Starts with	^demo	demos, demonstration	my demo
\$	Ends with	demo\$	my demo	demonstration
?	Zero or one times (greedy)	demos?123	demo123, demos123	demoA123
??	Zero or one times (lazy)			
*	Zero or more times (greedy)	goo*gle	gogle, goooogle	goggle
*?	Zero or more times (lazy)			
+	One or more times (greedy)	goo+gle	google, goooogle	gogle, goggle
+?	One or more times (lazy)			
{n}	n times exactly	w{3}	www	w, ww
{n,m}	from n to m times	a{4, 7}	aaaa, aaaaa, aaaaaa, aaaaaaa	aaaaaaaa, aaa, a
{n,}	at least n times	go{2,}gle	google, gooogle, goooogle	ggle, gogle
()	Group	^(demo example)[0-9]+	demo1, example4	demoexample2
(?:)	Passive group (Useful for filters)			
\	Escape	AU\\$10	AU\$10, AU\$100	AU10, 10
\s	White space			
\S	Non-white space			
\d	Digit character			
\D	Non-digit character			
\w	Word			
\W	Non-word (e.g. punctuation, spaces)			

Outline

- **Introduction to Database**
- **How to install MySQL**
- **How to create Database**
- **How to use SQL for data analysis**
- **Kahoot Quiz**
- **Summary**





DATA ANALYST SQL QUERIES

Basics

- SELECT
- WHERE
- HAVING
- GROUP BY
- ORDER BY
- HAVING

Intermediate

- BETWEEN
- LIKE
- NULL
- IN
- OFFSET
- COALESE

Joins

- LEFT JOIN
- RIGHT JOIN
- INNER JOIN
- SELF JOIN

ADVANCED

- WINDOWS
- FUNCTIONS
- RANK
- PIVOT
- CTE

